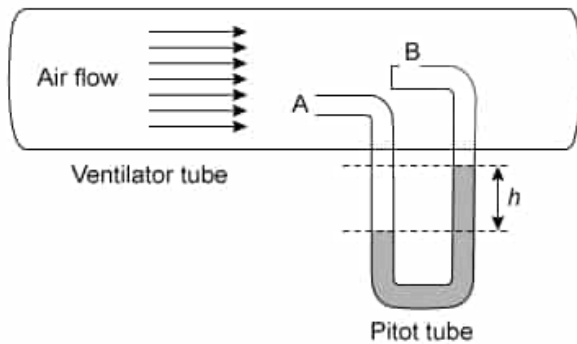




A pitot tube is used to measure the velocity of air flow delivered to a patient from a mechanical ventilator. Opening A is at the front of the tube and opening B is on the side of the tube. The fluid between these two openings blocks the air flow, causing the air velocity at opening A v_A to equal zero. Which of the following expressions is proportional to the height h of the fluid inside the pitot tube? (Note: Assume the air behaves like an ideal fluid; v_A and v_B are the air velocities at openings A and B, respectively.)

- ☐ A. $\frac{1}{v_B}$ [22%]
- ☐ B. $\sqrt{v_B}$ [29%]
- ☐ C. v_B [12%]
- ✓ ☒ D. v_B^2 [35%]

Correct

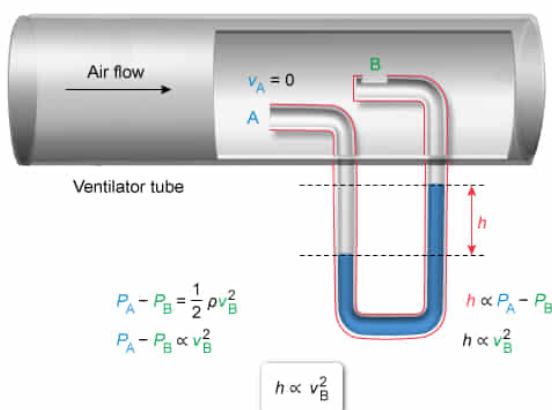
35% answered correctly

Explanation:

Pitot tube to measure air velocity

- P_A Pressure at opening A
- ρ Density of air
- v_A Velocity at opening A
- P_B Pressure at opening B
- v_B Velocity at opening B
- h Height of fluid

$$P_A + \frac{1}{2} \rho v_A^2 = P_B + \frac{1}{2} \rho v_B^2$$



©UWorld

Bernoulli's equation describes the flow of an ideal fluid within a pipe between two

points, A and B. Bernoulli's Principle is a special case of Bernoulli's Equation in which the height is constant and the **potential energy** terms cancel:

$$P_A + \frac{1}{2}\rho v_A^2 = P_B + \frac{1}{2}\rho v_B^2$$

A **pitot tube** measures **air velocity** based on this equation. The pitot tube is open at both ends and is curved to hold fluid between the openings. Opening A is at the end of the tube facing into the direction of air flow. Opening B is on the side of the tube. Because air does not flow into opening A, the air velocity v_A equals zero. Therefore, the equation above simplifies to:

$$P_A = P_B + \frac{1}{2}\rho v_B^2$$

Rearranging this equation shows the pressure difference is **directly proportional** to the square of v_B :

$$P_A - P_B = \frac{1}{2}\rho v_B^2$$

$$P_A - P_B \propto v_B^2$$

In this question, the fluid inside the pitot tube is exposed to P_A on one side and P_B on the other. Similar to a **simple fluid barometer**, the hydrostatic pressure P_h of the fluid equals the product of its density ρ_f , gravitational acceleration g , and height h :

$$P_h = \rho_f g h$$

$$h \propto P_h$$

In this case, P_h equals the difference between P_A and P_B :

$$P_h = P_A - P_B$$

$$h \propto P_A - P_B \propto v_B^2$$

Therefore, h is proportional to the square of v_B :

$$h \propto v_B^2$$

(Choices A, B, and C) The height of the fluid is proportional to the kinetic energy of the flowing air, which is proportional to the square of the air velocity at opening B, not the inverse of the velocity, the square root of the velocity, or the velocity itself.

Educational objective:

A pitot tube measures the velocity of air flow. The height of the fluid in the tube is proportional to the square of the air velocity at the opening on the side of the tube.